Fraud Screening Agent

An agent that screens card transactions and flags likely fraud for review.

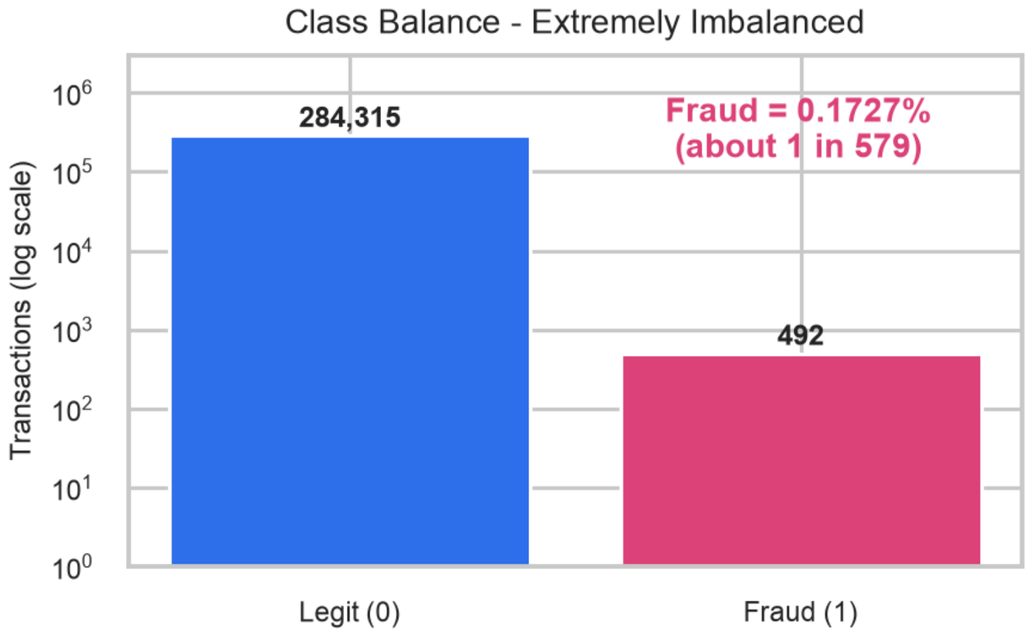
Irish Rosita · SoSe26 · Group A

Environment (Russell & Norvig)

Property	Classification	Reasoning
Observable	**Partially observable**	Sees only the transaction fields, not who is really holding the card.
Deterministic / Stochastic	**Stochastic**	Similar-looking transactions can be fraud or legit.
Episodic / Sequential	**Episodic**	Each transaction is scored on its own.
Static / Dynamic	**Dynamic**	Fraud tactics and spending patterns keep changing.
Discrete / Continuous	**Mostly discrete**	Actions are approve or flag. Inputs like amount are continuous.
Single / Multi-agent	**Multi-agent**	Fraudsters adapt to what the agent catches.

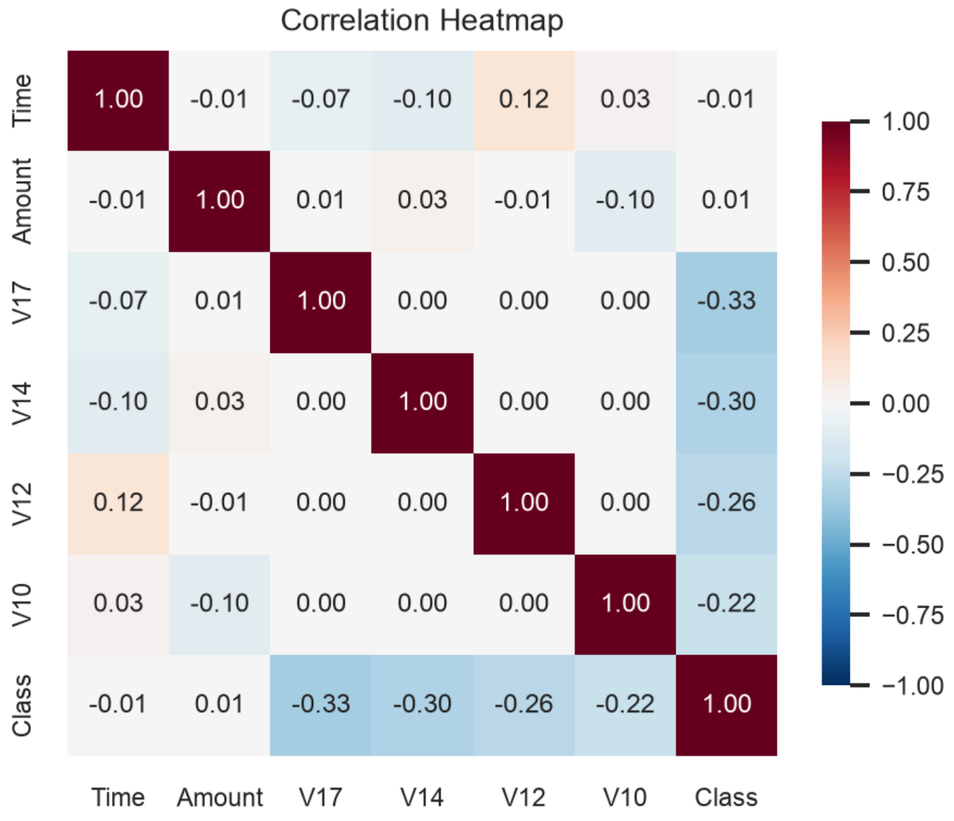
Data

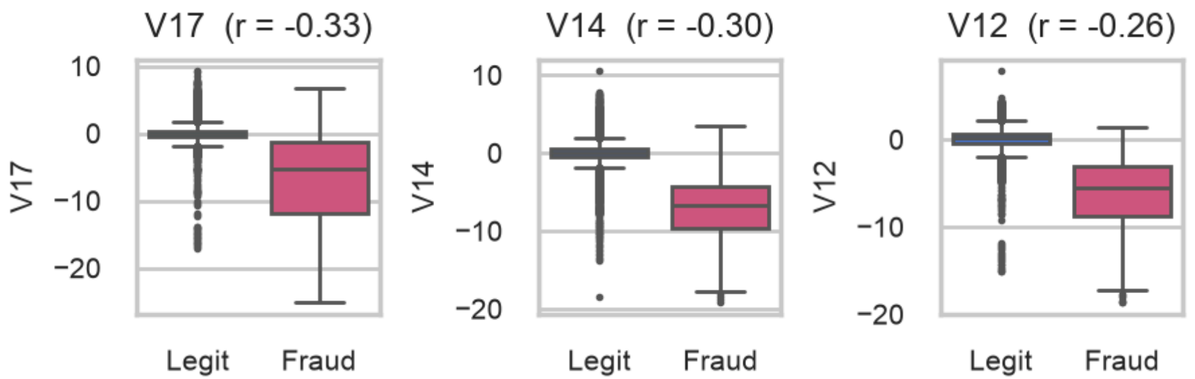
Source: Kaggle, ULB Credit Card Fraud Detection (kaggle.com/datasets/mlg-ulb/creditcardfraud)
Represents: 284,807 real European card transactions over 48 hours, each labeled legit or fraud.
Fields: Time, Amount, V1 to V28 (anonymized), Class (0 = legit, 1 = fraud).
In use: a live stream of new transactions with the same fields.
Problems: only 0.17% fraud (492), 1,081 duplicate rows, anonymized features.

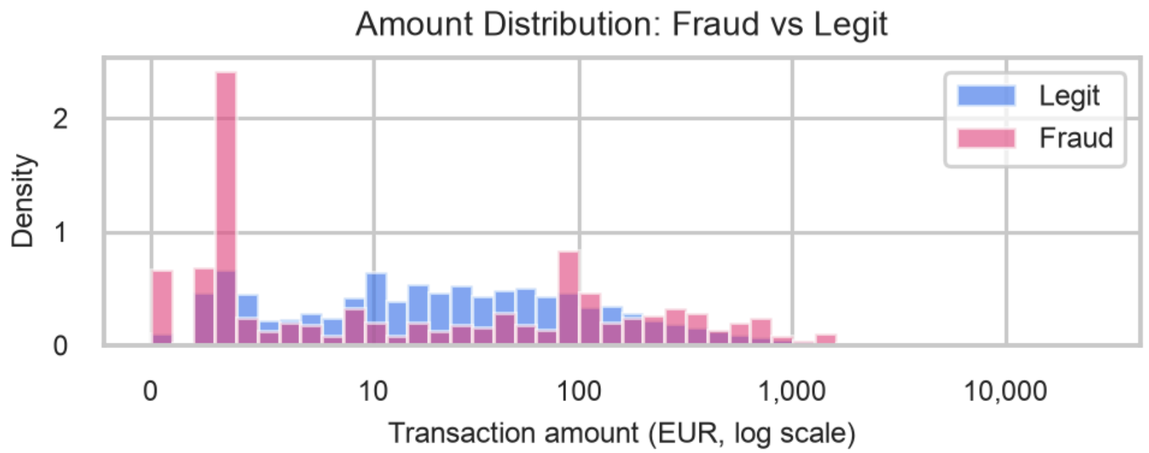


Time	V1	V2	V14	V17	Amount (EUR)	Class
0	-1.36	-0.07	-0.31	0.21	149.62	0
0	1.19	0.27	-0.14	-0.11	2.69	0
406	**-2.31**	**1.95**	**-4.29**	**-2.83**	**0.00**	**1**
472	**-3.04**	**-3.16**	**-1.69**	**0.60**	**529.00**	**1**

Data Inspection



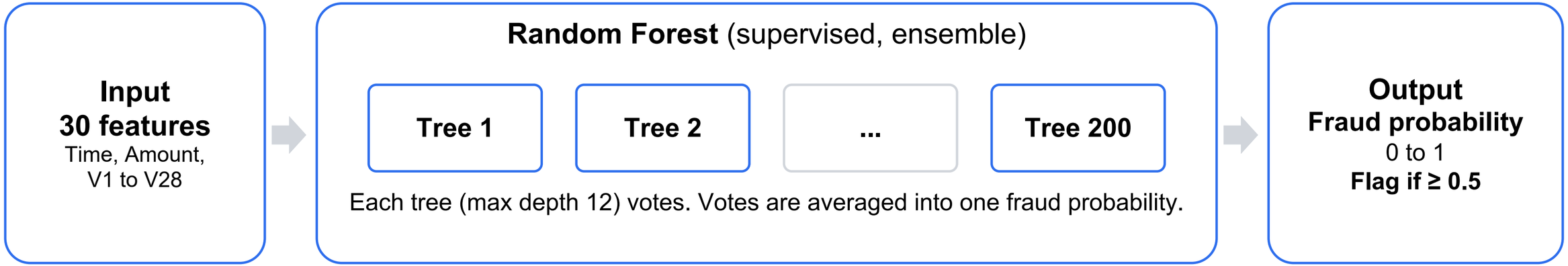




V17, V14 and V12 separate fraud best (much lower values). 36.8% of frauds are EUR 1 or less, vs 10.7% of legit.

Model: Random Forest

Supervised learning on labeled transactions. An ensemble of decision trees that vote.



Major parameter	Value	What it does
n_estimators	200	Number of trees in the forest
max_depth	12	How deep each tree can grow
min_samples_leaf	2	Smallest group of rows allowed in a leaf (limits overfitting)
class_weight	"balanced"	Missing a fraud costs more, to offset the 0.17% fraud rate
Threshold	0.5	Probability needed to flag a transaction
Train / test split	75 / 25	213,605 train rows, 71,202 test rows (stratified)

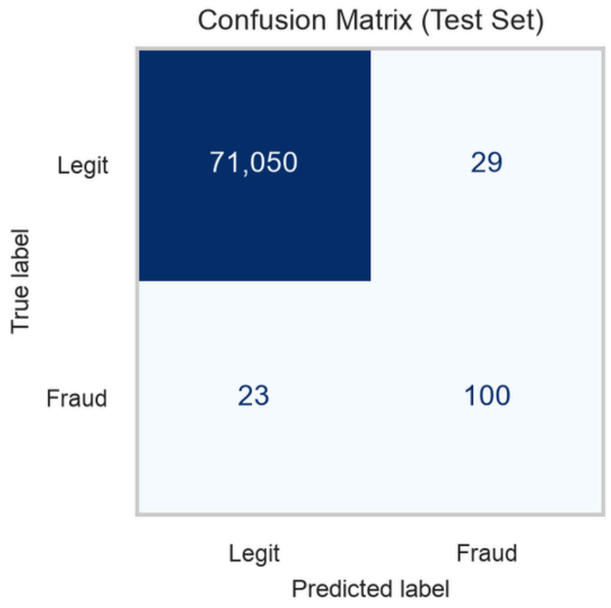
Results

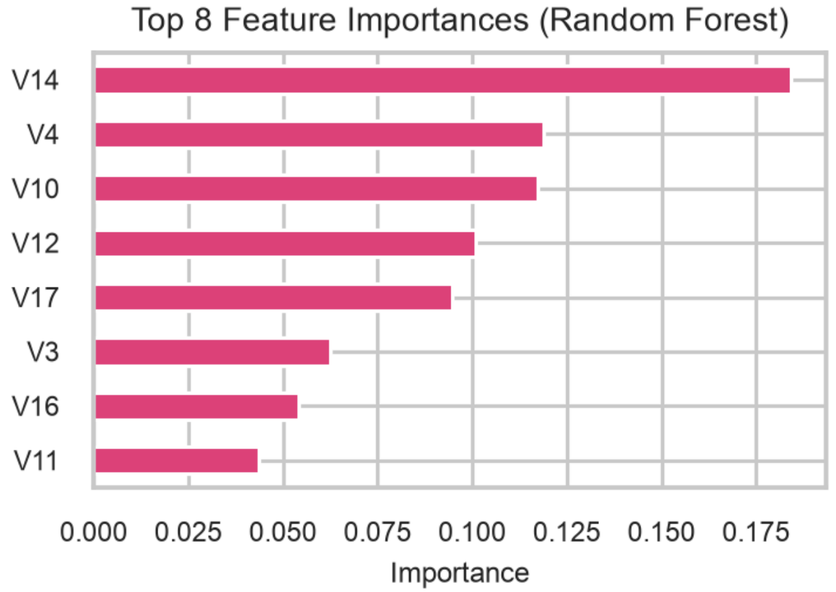
81.3%
Recall (fraud caught)

77.5%
Precision (alerts that were fraud)

0.794
F1 score

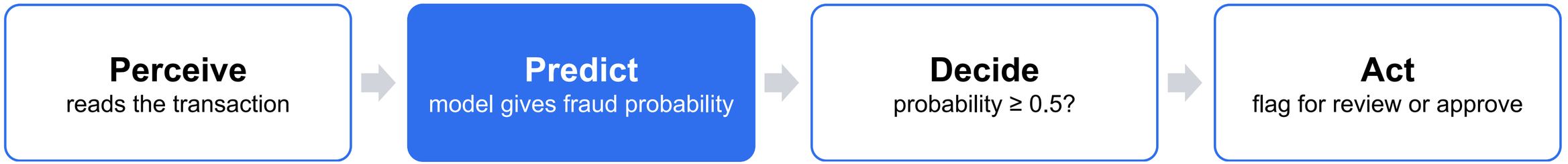
0.976
ROC-AUC





Accuracy is misleading. 99.93% sounds great, but always saying "legit" scores 99.83%.
Missed 23 frauds, wrongly flagged 29 real transactions.
To improve: a 0.3 threshold lifts recall to 87.0% but adds false alarms (86). Tune settings and add more fraud examples.

How the Model Serves the Agent



Understands
Which patterns look like fraud, learned from 284,807 past transactions. Most important features: V14, V4, V10.

Decides
Flag or approve each transaction. The threshold trades missed fraud against false alarms.

Limits
Cannot say why (features are anonymized), cannot see intent, and does not learn after training.

Agent type: a learning agent while training, a simple reflex agent when running (if score ≥ 0.5, flag).